

Due: Fri, 25 Sep 2026 17:00

100 Points Possible

Attempt 1

In Progress

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Unlimited Attempts Allowed
Available: 31 Aug 2026 9:00 until 2 Oct 2026 17:00

Details

Assessment Overview - Assignment 2

Assessment Type	Individual Written Assignment
Purpose	The purpose of this assignment is to explore, implement, and compare different multi-modal data fusion techniques.
Weighting	20%
Due Date	Friday 5:00 pm, Week 8
Submission Method	Online
Time Limit & Attempts	You can submit any time before the due date/time. You are allowed unlimited submissions before the due date/time.
Academic Integrity Checklist	Do ✓ Adhere to the university's policies on academic integrity. ✓ Complete your work independently. Do NOT ✗ Post/share code or complete solutions in Discussion threads/on the Internet etc. ✗ Give/show your answers to others.
Feedback	Returned Work - Within 15 university working days (3 weeks) after the due date.

Assignment Tasks

Objective

The purpose of this assignment is to explore, implement, and compare different multi-modal data fusion techniques in a typical ML pipeline. The goal of this analysis is to develop models that can classify the type of activity based on sensor readings.

Your dataset is the following: [MHEALTHDATASET_MISSING.zip \(https://learn.adelaide.edu.au/courses/30821/files/6201739?wrap=1\)](https://learn.adelaide.edu.au/courses/30821/files/6201739?wrap=1).  [\(https://learn.adelaide.edu.au/courses/30821/files/6201739/download?download_frd=1\)](https://learn.adelaide.edu.au/courses/30821/files/6201739/download?download_frd=1)

This dataset is an adapted version of the MHEALTH dataset by Oresti Banos, Rafael Garcia, and Alejandro Saez, obtained from the UCI Machine Learning Repository (DOI: [10.24432/C5TW22 \(https://archive.ics.uci.edu/dataset/319/mhealth+dataset\)](https://archive.ics.uci.edu/dataset/319/mhealth+dataset)), licensed under CC BY 4.0. The dataset has been modified for teaching purposes.

Task details

The following tasks should be completed and submitted as a technical report. See the submission guidelines at the bottom of this page for how you should structure your report.

Task 1: Data Exploration (20 Marks)

- Download and describe the adapted MHEALTH dataset.
- Perform exploratory data analysis (EDA) using appropriate summary statistics and visualisations.
- Investigate how selected sensor measurements vary across activities and/or participants.
- Transform the raw files into a structured format suitable for subsequent analysis.

Task 2: Missing Data and Preprocessing (20 Marks)

- Identify and quantify missing values within the dataset.
- Select, justify, and implement an appropriate missing-data strategy.
- Apply appropriate preprocessing to the sensor measurements where required, such as interpolation, standardisation, or normalisation.
- Segment the sensor data into appropriate time windows and extract suitable features from each window.
- Ensure that preprocessing avoids inappropriate mixing of observations between participants or activities.

Task 3: Multimodal Fusion and Modelling (35 Marks)

- Select at **least three sensor sources, representing at least two different sensor modalities** for activity classification.
- Establish appropriate unimodal baseline models using the selected modalities independently.
- Implement and evaluate **two different multimodal fusion strategies**, such as:
 - data-level (early) fusion;
 - feature-level (mid) fusion;
 - decision-level (late) fusion.
- *Note that you may wish to implement different versions of a particular fusion method or research other options.*
- Ensure that the same training and testing strategy is used across experiments to enable fair comparison.

Task 4: Evaluation and Critical Discussion (25 Marks)

- Evaluate all models using appropriate classification metrics, including accuracy, precision, recall and F1-score.
- Use confusion matrices and/or other appropriate visualisations to investigate model performance.
- Compare unimodal and multimodal approaches and determine whether fusion improved classification performance.
- Compare the advantages and limitations of the implemented fusion approaches.
- Discuss the computational implications of incorporating additional sensors and fusion stages.
- Discuss relevant ethical considerations.
- Discuss limitations of the methods and identify potential improvements.

Submission Guidelines

- Report format: PDF.
- Report structure:
 - You should write in a technical report style with clear headings for each task.
 - While you should use the tasks as section headings, a strong submission (i.e. D and HD level) will demonstrate connections across sections.
 - You should write in full sentences with referencing where appropriate.
 - The document should be well formatted.
- Maximum length: 10 pages (excluding figures, references, and appendices).
- Figures and tables should be well formatted with each being directly referenced and discussed in the text. This means you should be considering how relevant each inclusion is to the task and/or supporting text.
- As a rough guide, most submissions will include between 6 and 8 figures across all tasks, though quality and relevance matter more than quantity.
- Code does not need to be included as it likely is quite long. However, you should keep all your code as we may require review of it if there are any issues during the marking.
- Read and follow the information regarding assessments [here](https://learn.adelaide.edu.au/courses/30821/pages/additional-course-administration-information) (<https://learn.adelaide.edu.au/courses/30821/pages/additional-course-administration-information>).

View Rubric

Assignment 2 (1)		
Criteria	Ratings	Points

Criteria	Ratings					Points
Data Exploration view longer description	High Distinction Dataset is described thoroughly, including its structure, participants, activities, sensor sources and measurements. EDA is insightful and goes beyond surface-level summary statistics, using appropriate visualisations to reveal and interpret meaningful patterns in sensor measurements across activities and/or participants. The raw data are transformed into a clear, consistent and analysis-ready structure, with relevant identifiers and information preserved for subsequent analysis. Choices made during exploration and data restructuring are clearly explained and demonstrate a strong understanding of the dataset. 20 to >16 pts	Distinction Dataset is described clearly and accurately, covering the main aspects of its structure, participants, activities, sensors and measurements. EDA uses appropriate summary statistics and visualisations to investigate relevant patterns across activities and/or participants, with clear interpretation of the main findings. The raw data are transformed into an appropriate structured format suitable for subsequent analysis, with only minor omissions or limitations in the depth of exploration or explanation. 16 to >14 pts	Credit Dataset is described accurately, with most relevant aspects of its structure, participants, activities and sensor measurements identified. EDA includes appropriate summary statistics and visualisations and provides some investigation of differences across activities and/or participants, although interpretation is limited or primarily descriptive. The raw data are transformed into a usable structured format, though some aspects of the transformation or resulting structure may be insufficiently explained. 14 to >12 pts	Pass Provides a basic but generally accurate description of the dataset and its main components. EDA includes some relevant summary statistics and visualisations, but exploration of sensor measurements across activities and/or participants is limited, superficial, or weakly interpreted. The raw data are transformed sufficiently to support subsequent analysis, although the resulting structure or transformation process may contain notable weaknesses or be poorly explained. 12 to >9 pts	Fail Dataset description is absent, substantially incomplete, or demonstrates significant misunderstanding of the data. EDA is minimal, inappropriate, or contains significant errors, with little meaningful investigation of sensor measurements across activities or participants. The raw data are not transformed into a suitable structure for subsequent analysis, or substantial errors in the transformation compromise the integrity or usability of the data. 9 to >0 pts	FNS Fail No Submission /20 pts
Missing Data and Preprocessing view longer description	High Distinction Missing values are comprehensively identified and quantified, with relevant patterns of missingness investigated. The selected missing-data strategy is appropriate to the sensor data, clearly justified, and correctly implemented. Sensor preprocessing is appropriate and well motivated, with transformations such as interpolation, standardisation or normalisation applied where required. Time-series segmentation is carefully designed and justified, and suitable features are extracted from each window to support subsequent modelling. Preprocessing is technically rigorous and explicitly avoids inappropriate mixing of observations between participants or activities. 20 to >16 pts	Distinction Missing values are clearly identified and quantified, with some investigation of relevant missingness patterns. An appropriate missing-data strategy is justified and correctly implemented. Suitable preprocessing, windowing and feature extraction are applied to the sensor measurements, with clear reasoning for the main methodological choices. Participant and activity boundaries are appropriately handled, with only minor limitations in the depth of justification or preprocessing design. 16 to >14 pts	Credit Missing values are correctly identified and quantified, and a reasonable missing-data strategy is implemented with some justification. Appropriate preprocessing, windowing and feature extraction are performed, although choices are primarily conventional or only partially justified. Participant and activity boundaries are generally respected, but some aspects of the preprocessing pipeline lack methodological detail or careful consideration. 14 to >12 pts	Pass Missing values are identified and a basic strategy is used to address them, but investigation of missingness and justification of the chosen approach are limited. Basic preprocessing, segmentation and feature extraction are implemented sufficiently to produce data suitable for modelling, although important choices are weakly justified or contain notable limitations. There is some consideration of participant and activity boundaries, but the approach may present minor risks of inappropriate mixing or leakage. 12 to >9 pts	Fail Missing data are inadequately identified or handled, or the selected strategy is inappropriate or incorrectly implemented. Preprocessing, windowing or feature extraction is absent, substantially incomplete, or contains significant methodological errors. Participant or activity boundaries are not appropriately considered, resulting in inappropriate mixing, data leakage, or processed data that are unsuitable for reliable subsequent modelling. 9 to >0 pts	FNS Fail No Submission /20 pts
Multimodal Fusion and Modelling view longer description	High Distinction Sensor sources and modalities are selected thoughtfully and clearly justified in relation to the activity-classification task. Appropriate unimodal baseline models are established and provide meaningful reference points for assessing multimodal performance. Two distinct fusion strategies are correctly and effectively implemented, with the fusion mechanisms and modelling choices clearly explained and well motivated. The experimental design is rigorous and consistent across unimodal and multimodal models, enabling valid and fair comparison while avoiding data leakage. The overall modelling approach demonstrates strong technical understanding of multimodal fusion and its application to sensor-based activity classification. 35 to >29 pts	Distinction Appropriate sensor sources spanning the required modalities are selected and well justified. Suitable unimodal baseline models are implemented, and two distinct multimodal fusion strategies are correctly applied and clearly explained. A consistent training and testing strategy is used across experiments, enabling meaningful comparison between models. The modelling approach is technically sound, with only minor limitations in justification, implementation, or experimental design. 29 to >26 pts	Credit The required sensor sources and modalities are appropriately selected, with some justification. Unimodal baseline models and two valid multimodal fusion strategies are implemented and generally explained correctly. Training and testing procedures are broadly consistent across experiments, allowing reasonable comparison, although some methodological choices lack justification or the fusion approaches demonstrate limited depth. 26 to >23 pts	Pass The minimum sensor and modality requirements are met, and unimodal baselines and two fusion approaches are attempted. The implementations are sufficient to demonstrate the basic principles of multimodal fusion, but sensor selection, modelling choices, or fusion strategies are weakly justified or contain notable limitations. A broadly comparable training and testing strategy is used, although weaknesses in experimental consistency may limit the strength of the comparison. 23 to >18 pts	Fail The required sensor sources, modalities, unimodal baselines, or two distinct fusion strategies are not adequately implemented. Fusion methods may be incorrectly applied or demonstrate substantial misunderstanding of multimodal modelling. Training and testing procedures are inconsistent or contain significant methodological errors, preventing reliable comparison between unimodal and multimodal approaches. 18 to >0 pts	FNS Fail No Submission /35 pts
Evaluation and Critical Discussion view longer description	High Distinction Model performance is evaluated comprehensively using accuracy, precision, recall and F1-score, supported by appropriate confusion matrices and/or other informative visualisations. Results are critically interpreted to identify meaningful performance patterns and determine whether and where multimodal fusion improves upon unimodal approaches. The advantages and limitations of the implemented fusion strategies are compared insightfully and supported by experimental evidence. Computational implications of additional sensors and fusion stages are considered thoughtfully. Ethical considerations are relevant and specific to the use of multimodal sensor data for activity classification. Limitations of the data, methods and evaluation are critically examined, with well-motivated and technically appropriate improvements proposed. 25 to >21 pts	Distinction Model performance is evaluated thoroughly using the required classification metrics and appropriate visualisations. Unimodal and multimodal approaches are clearly compared, with good interpretation of whether fusion improves classification performance. The advantages and limitations of the fusion approaches and their computational implications are discussed clearly. Relevant ethical considerations and methodological limitations are identified, with sensible potential improvements proposed. Discussion is well supported by the experimental results, with only minor limitations in depth or critical analysis. 21 to >18 pts	Credit The required classification metrics are correctly reported and supported by suitable performance visualisations. Unimodal and multimodal models are compared and the effect of fusion on performance is discussed, although interpretation is primarily descriptive. Relevant advantages and limitations of the fusion approaches, computational considerations, ethical issues and methodological limitations are addressed, but discussion lacks depth or connection to the experimental findings in some areas. Reasonable improvements are identified. 18 to >16 pts	Pass Basic evaluation is provided using most or all of the required classification metrics and at least some appropriate performance visualisation. Unimodal and multimodal results are compared, but interpretation is limited and largely restates reported results. Advantages, limitations, computational implications, ethical considerations and potential improvements are addressed at a basic or generic level, with limited connection to the implemented methods or experimental evidence. 16 to >12 pts	Fail Evaluation is incomplete, substantially incorrect, or uses inappropriate metrics, and model performance is not meaningfully investigated. Comparisons between unimodal and multimodal approaches are absent, unsupported, or inconsistent with the reported results. Discussion of fusion trade-offs, computational implications, ethical considerations, limitations and potential improvements is substantially incomplete or demonstrates significant misunderstanding of the task. 12 to >0 pts	FNA Fail No Submission /25 pts